

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Second Semester B.Tech Degree Regular and Supplementary Examination June 2023 (2019 Scheme)

Course Code: MAT 102**Course Name: VECTOR CALCULUS, DIFFERENTIAL EQUATIONS AND TRANSFORMS
(2019 SCHEME)**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all Questions. Each question carries 3 Marks*

Marks

- 1 Find the parametric equation of the tangent vector of the curve $\vec{r}(t) = t^2 \hat{i} + 2t^3 \hat{j} + 3t \hat{k}$ at $t = 1$. (3)
- 2 Find the directional derivative of $f(x, y) = xe^y$ at (1,1) in the direction of the vector $\hat{i} - \hat{j}$. (3)
- 3 Use Green's theorem to evaluate $\oint_C xdy - ydx$, where C is the circle $x^2 + y^2 = 4$. (3)
- 4 Determine whether the vector field $\vec{F} = 4(x^3 - x)\hat{i} + 4(y^3 - y)\hat{j} + 4(z^3 - z)\hat{k}$ is free of sources and sinks. If not locate them. (3)
- 5 Show that the functions $x, x \ln x$ are linearly independent. (3)
- 6 Solve the differential equation $y'' + 4y' + 2.5y = 0$. (3)
- 7 Find the Laplace transform of $\sin 4t \cos 3t$. (3)
- 8 Find the Laplace transform of $e^{-3t}u(t - 1)$. (3)
- 9 Determine the Fourier sine Transform of $f(x) = 3x, 0 < x < 6$. (3)
- 10 Find the Fourier sine integral of $f(x) = \begin{cases} \sin x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$ (3)

PART B*Answer one full question from each module, each question carries 14 marks***Module I**

- 11 a Find the divergence and curl of the vector field $\vec{F}(x, y, z) = zy\hat{i} + y^2x\hat{j} + yz^2\hat{k}$. (7)
- b Show that $\vec{F} = (\cos y + y \cos x)\hat{i} + (\sin x - x \sin y)\hat{j}$ is a conservative vector field. Hence find a potential function for it. (7)

OR

- 12 a Find the work done by the force $\vec{F} = xy\hat{i} + yz\hat{j} + xz\hat{k}$ on a particle that moves (7)
 along the curve $C: \vec{r}(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ where $0 \leq t \leq 1$.
- b Show that $\int_C (3x^2e^y dx + x^3e^y dy)$ is independent of the path and hence evaluate (7)
 the integral from (0,0) to (3,2).

Module II

- 13 a Using Green's theorem, evaluate the line integral $\int_C (xy + y^2) dx + x^2 dy$ (7)
 where C is bounded by $y = x$ and $y = x^2$ and positively oriented.
- b Evaluate the surface integral $\iint_\sigma z^2 dS$, where σ is the portion of the cone (7)
 $z = \sqrt{x^2 + y^2}$ between the planes $z=1$ and $z=3$.

OR

- 14 a Evaluate $\iint_S \vec{F} \cdot \hat{n} dS$, where S is the surface of the cylinder $x^2 + y^2 = 4, z =$ (7)
 $0, z = 3$ where $\vec{F} = (2x - y)\hat{i} + (2y - z)\hat{j} + z^2\hat{k}$.
- b Apply Stoke's theorem to evaluate $\int_C (x - y)dx + (y - z)dy + (z - x)dz$ (7)
 where C is the boundary of the portion of the plane $x + y + z = 1$ in the first octant.

Module III

- 15 a Solve the Cauchy -Euler differential equation $(x^2 D^2 - 3xD + 10)y = 0$. (7)
- b Solve the initial value problem $y'' - 2y' + 5y = 0, y(0) = -3, y'(0) = 1$. (7)

OR

- 16 a By the method of undetermined coefficients, solve $y'' + y' - 2y = \sin x$. (7)
- b Using method of variation of parameters solve $\frac{d^2 y}{dx^2} + 4y = \sec 2x$. (7)

Module IV

- 17 a Find the Laplace transform of $\cos^2 t$. (7)
- b Find the inverse Laplace transform of $\frac{3s+2}{(s-1)(s^2+1)}$. (7)

OR

- 18 a Using Laplace transform solve $y'' + 5y' + 6y = e^{-2t}$ given that (7)
 $y(0) = y'(0) = 1$
- b Find the inverse Laplace transform of $\frac{s}{(s^2+a^2)^2}$ using convolution. (7)

Module V

- 19 a Find the Fourier transform and integral representation of (7)
 $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & \text{otherwise} \end{cases}$. Hence show that $\int_0^\infty \frac{\sin w}{w} dw = \pi/2$

- b Find the Fourier sine transform and inverse transform of (7)
 $e^{-ax}, x > 0 \quad a > 0$

OR

- 20 a Find the complex Fourier transform of $f(x) = \begin{cases} \sin x, & |x| \leq a, a > 0 \\ 0, & |x| > a \end{cases}$ (7)

- b Find the Fourier transform and integral representation of (7)

$$f(x) = \begin{cases} 1 - x^2, & |x| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$
